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MILITARY SPECIFICATION

PUMPING ASSEMBLY, FLAMMABLE LIQUID, BULK TRANSFER, TRAILER MOUNTED A/M32R-22 DIESEL ENGINE, 600 GPM

1. SCOPE

1.1 This specification covers a trailer-mounted, diesel-engine-driven pumping assembly, designated A/M32R-22.

2. APPLICABLE DOCUMENTS

*2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-514	Plate, Identification, Instruction and Marking, Blank
MIL-T-5021	Tests, Aircraft and Missile Welding Operators, Qualifications
MIL-C-5424	Cable, Steel (Corrosion-Resisting), Flexible, Preformed (For Aeronautical Use)
MIL-G-5572	Gasoline, Aviation: Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-H-8794	Hose, Rubber, Hydraulic, Fuel and Oil Resistant
MIL-E-10062	Engine, Preparation For Shipment and Storage of
MIL-C-10387	Coupling, Clamp, Pipe, with Bolts and Synthetic Rubber Gasket, For Grooved Pipe and Tube
MIL-P-10388	Pipe Fittings, One or More Ends Grooved
MIL-E-11276	Engine, Diesel, Industrial, Medium- and High-Speed, General Specification For
MIL-R-11461	Rod, Ground (With Attachments)
MIL-N-26978	Nozzle, Fuel and Oil Servicing, Over-The-Wing, Shockproof, Type MD-1 and MD-2
MIL-C-27487	Coupling Halves, Quick Disconnect-Cam-Locking Type
MIL-N-27506	Hose Assembly, Rubber GRK-6/E32R-1
MIL-H-27516	Hose Assembly, Rubber GRK-7/E32R-1
MIL-S-27661	Strainer, Sediment, Fuel Servicing Unit, 600 GPM
MIL-C-46722	Chassis, Trailer, 1 to 5 Ton, 4 Wheel, General Purpose
MIL-V-58039	Valves, Gate, Rising Stem, Double Acting, Aluminum

FSC 4320

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STANDARDS

Federal

FED-STD-595 Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-276	Impregnation of Porous Nonferrous Metal Castings
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MS33586	Metal, Definition of Dissimilar

Air Force-Navy Aeronautical

AN2552	Receptacle - External Power, 28 Volt DC

DRAWINGS

Air Force

50C24046	Coupling Male, 6-Inch Air Duct
54C6211	Clamp, Hold Down Equipment, Trailer Chassis, Assembly of

PUBLICATION

Air Force Bulletin

518	Cargo, Aircraft Compartment Dimensional Data

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Standards Association

S1.4	General Purpose Sound Level Meters
S1.6	Preferred Frequencies for Acoustical Measurements
Z24.10	Octave Band Filter Sets for the Analysis of Noise and Other Sounds

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

*3.2 Components. The pumping assembly shall consist of the following major components:

Description	See Requirement
Trailer	3.7.1
Base	3.7.2
Engine	3.7.3
Pump	3.7.4
Line strainer	3.7.5
Connectors	3.7.6
Adapters	3.7.7
Control box	3.7.8
Ground rod	3.7.9
Gate valve assembly	3.7.10.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Protective treatment. When materials are used in the construction of the pumping assembly that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with use, age, or extremes of climatic and environmental conditions shall be avoided.

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3.4.2 Fungusproof materials. Materials which are not nutrients for fungi shall be used to the greatest practicable extent. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent acceptable to the procuring activity.

3.4.3 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service. All components that come into contact with the fuel shall be fabricated of nonferrous metals or 300 series stainless-steel.

3.4.3.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.3.2 Bonding. Metal parts shall be specifically bonded together to prevent the possibility of sparking.

3.4.3.3 Impregnation of castings. Aluminum castings may be impregnated in accordance with MIL-STD-276 to prevent minor weeping.

3.4.4 Elastomers. Elastomeric materials used shall be compatible with fuels conforming to MIL-G-5572 and MIL-T-5624 and shall withstand -40° to +130°F temperatures.

3.4.5 Wood. Unless specifically required herein or approved by the procuring activity, wood shall not be used.

3.5 Design and construction. The pumping assembly shall be so designed and constructed that parts will not work loose in service and shall be inherently capable of withstanding the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The pumping assembly shall be designed and constructed for maximum ease and safety of operation. Pins, brackets, collars, dust covers, dust plugs, et cetera, which are detachable in normal operation shall be attached to the assembly using 1/16-inch-diameter cable in accordance with MIL-C-5424.

3.5.1 Reliability. The pumping assembly shall have an MTBF of not less than 150 hours with a reliability of not less than .9933 for a 1-hour mission at 0.9 confidence. (See 6.3.1 through 6.3.1.2 and 6.4.)

3.5.2 Component accessibility. The pumping assembly shall be so designed that all components may be readily removed, serviced, and operated. Any design which renders operation unduly difficult under field conditions will not be acceptable. The pumping assembly shall be designed to insure ease of production, ease and rapid installation of equipment, and ease of general maintenance and repair.

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3.5.3 Wiring. The electrical wiring shall be vaporproof in accordance with current flammable tank vehicle industry practice. All wiring shall be enclosed in rigid, metal conduit or flexible conduit covered with polyvinylchloride firmly attached to the trailer. Unless in a vaportight enclosure, all switches and electrical devices shall be of the hermetically sealed type. The entire electrical system and all the components shall be designed to withstand a 11-psi positive or negative pressure without leaking. The electrical wire shall be a minimum of 16-gage wire with moisture-resistant thermoplastic insulation. Each wire shall be marked at the ends and in the middle. This marking shall correspond to a similar marking on the wiring schematic.

3.5.4 Piping. The piping layout shall be designed to facilitate maintenance on the pumping system and removal of components with minimum disassembly of piping. Groove-type couplings conforming to MIL-C-10387 shall be used to the maximum practical extent. Piping sections shall be easily removable for maintenance and repair. To the maximum practical extent, piping connections to major and minor components shall be of the groove type conforming to MIL-C-10387 and MIL-P-10388. The piping shall be designed to facilitate draining to the maximum practical extent. The large pipe sections and pipe sections which could be subject to excessive movement during operation shall be firmly mounted or braced. Pipe sections such as drain lines that could be damaged during operation of their valves shall be supported. Mitered bends will not be acceptable.

3.5.4.1 Tubing and conduit. Wherever possible, static pressure and gage-sensing tube lines shall be installed in groups and mounted on blocks which provide for separation of the lines and mounting of multiple lines in one place on any flat surface. Tubing and conduit shall be firmly supported such that vibration is held to a minimum and so that they do not vibrate against one another or any component of the pumping assembly.

3.5.5 Lubrication. Whenever practicable, lubricated-for-life components or components not requiring lubrication shall be used in the pumping assembly design. Grease seals shall be designed and located to provide maximum accessibility for inspection, service, and replacement and to prevent displacement, rupture, or other failure of seals due to excessive amounts of lubricant or lubricant pressure. Grease seals shall be so designed that pressure due to forced lubrication will not bind rotating parts. Lubricating grooves shall be provided as required to insure uniform lubrication of internal parts.

3.5.6 Orientation. The control box and pump discharge shall be on the right side of the pumping assembly when looking from the engine towards the pump.

3.5.7 Air transportability. The pumping assembly shall be air transportable in accordance with MIL-A-8421 and Bulletin 518 in Types C-123, C-124, C-130, C-133, and C-141 aircraft.

3.5.8 Handtools. The pumping assembly shall be designed and constructed in a manner that will allow general maintenance, assembly, and disassembly to be performed to the maximum practical extent with the use of common handtools.

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3.5.9 Welding. All welders employed in the fabrication of the pumping assembly shall be certified in accordance with class A, groups I and IV, of MIL-T-5021.

3.6 Performance

3.6.1 Environmental conditions

3.6.1.1 Temperature ranges

3.6.1.1.1 Storage temperatures. The pumping assembly shall not be damaged by storage in ambient temperatures ranging from -80° to +160°F for not less than 50 hours.

3.6.1.1.2 Operating temperatures. The pumping assembly shall be capable of performing as specified herein in ambient temperatures between -40° and +125°F. Starting in ambient temperatures between -40° and -20°F may be facilitated by external heat supplied through the heater adapter specified in 3.7.3.8.1 to raise the engine temperature to -20°F.

3.6.1.2 Humidity. The pumping assembly shall not be damaged by operation or storage in any relative humidity up to and including 100 percent, including conditions wherein condensation takes place in the form of water and frost.

3.6.1.3 Fungus. The pumping assembly shall not be damaged by exposure to moist fungus growth such as encountered in tropical and subtropical climates.

3.6.1.4 Salt atmosphere. The pumping assembly shall not be damaged by operation or storage in atmospheres containing salt-laden moisture such as encountered near bodies of salt water and in transportation on shipboard.

3.6.1.5 Dust. The pumping assembly shall not be damaged by operation or storage in atmospheres containing sand and dust particles as encountered in desert areas.

3.6.1.6 Rain. The pumping assembly shall not be damaged by operation or storage in rainfall as encountered in any locale. The pumping assembly shall not have areas that would trap and retain water from rainfalls.

3.6.2 Pumping capacity. The pumping assembly shall deliver 600 gpm at a total head of 130 psig when pumping grade JP-4 fuel in accordance with MIL-T-5624 with a specific gravity of at least 0.78 at a temperature of 60°F and a static suction life of 10 feet. The pumping assembly shall exhibit no instability in pressure, flow rate, or engine operation under any operating condition.

*3.6.3 Mobility. The pumping assembly shall be capable of being towed at 20 mph over paved highways, 10 mph over gravel roads, and 8 mph over Belgian-block pavement (see 6.3.2 and 6.3.3).

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3.6.4 Working and hydrostatic pressures. The pumping assembly shall withstand a working pressure of 150 psig and a hydrostatic pressure of 200 psig.

3.7 Details of components

3.7.1 Trailer. The trailer shall be a model MB-1 in accordance with MIL-C-46722; except that service brakes and automatic actuation of vehicle brakes, and the ten items, part No. 54C6211, shall not be provided. Parking brakes shall be furnished. Boltholes with nuts and bolts to attach the pumping assembly base to the trailer frame shall be provided.

3.7.2 Base. A base shall be provided for supporting, mounting, and retaining the components specified herein (except the trailer). The base shall be of aluminum alloy with lifting rings to facilitate removal of the base with components from the trailer. The base shall be provided with boltholes to permit bolting to the trailer frame. The base shall have sufficient structural integrity to prevent damaging stresses and forces from being transmitted to the components when lifted using the lifting rings and when attached to the trailer. Unless otherwise specified (see 6.2), the components shall be retained by removable-type fasteners such as nuts and bolts. The base dimensions shall not exceed those of the trailer frame.

3.7.3 Engine. The engine shall be a multiple-cylinder, liquid-cooled, diesel engine conforming to MIL-E-11276. The power and rotating speeds required of the engine to meet the performance requirements specified herein at sea level to 6,000 feet altitude at 104°F using grade JP-4 or JP-5 fuel conforming to MIL-T-5624 or diesel fuel shall not exceed the continuous rating as listed in the QPL for MIL-E-11276. The engine shall be complete with accessories as required by MIL-E-11276, except as modified or specified herein. The engine shall be rigidly mounted to the base by means of bolts. Vibration isolation may be provided between the engine and the base specified in 3.7.2.

3.7.3.1 Fuel tank. A 5-gallon fuel tank shall be provided. The necessary piping and valves shall be provided to permit automatic filling and maintaining a three-quarters filled condition using fuel and pressure from the discharge side of the pump. A valve shall be provided to permit manual control of this fuel flow.

3.7.3.1.1 Filler opening. A filler opening designed to receive a 2-inch overwing nozzle in accordance with MIL-N-26978 shall be incorporated into the fuel tank. The opening in the tank shall be located to prevent spillage or overflow from contacting hot components.

3.7.3.1.2 Filler cap. A filler cap, for connection to the filler opening, shall be provided and firmly retained during shipment and operation. The cap shall permit the tank to breathe, but shall prohibit the entrance of rain up to 45 degrees from the vertical. A 1/16-inch-diameter security cable in accordance with MIL-C-5424 shall be provided.

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3.7.3.1.3 Drain plug. A drain plug shall be provided to permit complete drainage of the fuel tank.

3.7.3.1.4 Fuel gage. A mechanically actuated fuel level gage graduated empty-to-full shall be provided.

3.7.3.2 Fuel transfer pump. A fuel transfer pump shall be provided. The pump shall be capable of supplying fuel to the engine from the fuel tank or an auxiliary supply tank such as a 55-gallon drum.

3.7.3.2.1 Fuel selector valves. Two three-way fuel selector valves shall be provided to permit engine operation from the fuel tank or an auxiliary fuel tank.

3.7.3.2.2 Auxiliary fuel lines. Two 10-foot-long hoses conforming to MIL-H-8794 and sized to permit adequate flow shall be provided, one on each auxiliary leg of the fuel selector valves. The free end of the hoses shall be provided with a reusable cover to prevent the entrance of foreign matter when not in use. Provisions shall be incorporated for readily storing, retaining, and removal of the hoses.

3.7.3.3 Priming system. The priming system shall permit controlled priming and shall be capable of utilizing bulk fluid and priming fluids containing upper cylinder lubricant.

3.7.3.4 Cranking system. The cranking system shall be electrical using a totally enclosed electrical cranking motor and a totally enclosed or brushless alternator of not less than a 15-amp rating.

3.7.3.5 Storage battery container. The storage battery container shall be fabricated using 300 series stainless-steel with a hinged cover with latches for holding it in a closed position and provisions for being held in the open position. The cover shall permit the enclosure to breathe. The batteries shall be rigidly retained in the container. Electrolyte checking and filling and battery removal and replacement shall be easily accomplished with the cover in the open position. There shall be a minimum of 1/4 inch clearance between the uppermost extremity of the battery and the cover and a minimum of 1 inch clearance between the battery terminals and the container sides. Provisions shall be furnished for catching and discharging spilled electrolyte without contaminating the pumping assembly.

3.7.3.6 Charging receptacle. A charging receptacle in accordance with AN2552 shall be provided. The receptacle shall be located in a relatively protected position and easily accessible for connection of heavy power cables. The cables from the battery to the receptacle shall be sized to accept normal fast-charging current without becoming hot. The receptacle shall be provided with a protective cover which is of the hinge type permanently attached or a removable cover with a 1/16-inch-diameter security cable in accordance with MIL-C-5424.

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3.7.3.7 Throttle. A manual throttle shall be provided for control of the engine speed through the engine governor for all pumping operations unless overridden by the safety controls. The throttle shall allow the operator to set the engine to any desired speed from idle to maximum rated speed as required to meet the performance requirements specified herein. The throttle shall provide for sufficient movement and smooth operation to allow the operator to select a previously determined operating speed within ±10 rpm without difficulty and also include a preset stop to prevent exceeding the overspeed control setting. A turn-to-lock mechanism shall be provided to allow the throttle to be locked at any desired setting.

3.7.3.8 Housing. The engine shall be completely enclosed. The housing shall provide protection from rain, snow, and sand; shall incorporate sound reduction features; shall include a heater adapter, manually operated shutter for the radiator; manually operated air vents for cooling air; and hinged sections using full-length, piano-type hinges with corrosion-resistant hinge pins to permit ready access for periodic and major maintenance. The hinged sections shall be provided with hold-open devices and hand-operated latches to hold them closed, and shall also be removable. The housing exterior shall be constructed of sheet steel with a minimum thickness of 16 gage (0.062 inch). All panels shall be integrally constructed and reinforced and constructed to prevent bending, warpage, or other types of damage due to vibration and other forces encountered during normal operation or transportation. All joints shall be tight. Sharp edges and corners which could be a hazard to operating or maintenance personnel will not be acceptable.

3.7.3.8.1 Heater adapter. The heater adapter shall be a 6-inch connector in accordance with Drawing 50C24046. The adapter shall be flush-mounted and provided with a cover or louver. If a cover is used, a 1/16-inch-diameter security chain in accordance with MIL-C-5424 shall be provided. The adapter shall be mounted to provide the best natural circulation around the engine to permit cold weather (-40°F) starting. The mounting position shall also allow adequate clearance for easy attachment of the heating duct.

3.7.3.9 Muffler. A spark- and flame-arrestor-type muffler with a rain cap shall be provided.

3.7.3.10 Oil drainage. The oil drain plug shall be replaced with a valve and appropriate length of flexible tubing connected to the valve discharge. The tubing shall be extended through the right side of the engine housing (when looking towards the pump) and shall be long enough to easily permit discharging into a container placed adjacent to the pumping assembly. The tubing shall be stored within the housing when not in use.

3.7.4 Pump. The pump shall be of the self-priming centrifugal type (see 6.3.4) close coupled to the diesel engine, and rigidly mounted to the base with provisions for aligning. Vibration isolation material may be provided between the pump mounts and the base specified in 3.7.2. The pump shall be of standard commercial manufacture utilizing wear- and corrosion-resistant parts and self-lubricating seals and bearings. A pump intermediate shall be provided upon which the pump impeller shall be mounted.

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The pump intermediate shall include a self-lubricating, self-adjusting mechanical type shaft seal, bearings which are capable of absorbing the thrust created by the pump, and a spline for connecting to the diesel engine. The pump shall be capable of producing the flows and pressures required to meet the requirements specified herein.

3.7.4.1 Pump housing. The pump housing shall be of aluminum alloy and capable of withstanding a hydrostatic pressure of 1-1/2 times the shutoff head developed by the pump when operating at maximum engine rpm. A drain plug shall be provided to permit complete drainage of the housing.

3.7.4.2 Dry operation. The pump shall operate at maximum rated pump speed for 5 minutes without appreciable wear or damage after being completely drained.

3.7.4.3 Efficiency. The pump shall have a minimum efficiency of 63 percent at rated output.

3.7.5 Line strainer. The line strainer shall be in accordance with MIL-S-27661, except the pipe connections shall be 6-inch connections and the working and test pressures shall conform to 3.6.4. The line strainer shall be located on the suction side of the pump.

3.7.6 Connectors

3.7.6.1 Inlet connector. The pumping assembly inlet connector shall be a 6-inch, female, class 1, quick disconnect with a dust plug in accordance with MIL-C-27487 and a 1/16-inch-diameter security cable in accordance with MIL-C-5424. This connector shall be at an angle from the horizontal towards the base and located to permit attachment of the 6-inch-diameter suction hose in accordance with MIL-H-27516 when operating from the Type MB-1 trailer or with the base resting on the ground. The location shall prevent the hose from rubbing against sharp corners or edges which could cause damage.

3.7.6.2 Outlet connector. The pumping assembly outlet connector shall be a 6-inch, male, class 1, quick disconnect with a dust cap in accordance with MIL-C-27487 and a 1/16-inch-diameter security cable in accordance with MIL-C-5424. This connector shall be at an angle of 45 degrees from the horizontal towards the base and located to permit attachment of a 6-inch-diameter discharge hose in accordance with MIL-H-27508. The location shall prevent the hose from rubbing against any sharp corners or edges which could cause damage, and from being subjected to any sharp bending.

3.7.7 Adapters. The following aluminum-alloy adapters with class 1, cam-locking, quick-disconnect coupling halves in accordance with MIL-C-27487 with a dust cap or plugs, with 1/16-inch-diameter security cable in accordance with MIL-C-5424, as applicable, shall be provided:

a. One 6-inch male to 4-inch female

b. One 6-inch female to 4-inch male.

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3.7.7.1 Adapter configuration. The configuration of the adapters shall be such that when they are connected to the 6-inch connectors, the 4-inch suction and discharge hose can be readily connected without sharp bends which might reduce the hose life or rub against any component or item which could damage the hose. This is applicable when the base is mounted on the trailer or positioned on the ground. A 6-inch dust cap and dust plug shall be bolted to the base and used as the means of storing and retaining the adapters.

3.7.8 Control box. A vaportight control box with a weathertight door shall be provided. The control box shall be either bolted or welded to the base. The cover shall be held closed by positive-type locking latches and shall have provisions for being held in the open position. The following gages, controls, et cetera, shall be provided and arranged for maximum convenience and in a logical manner. Only the electrical components shall be in the vaportight portion of the control box.

a. Engine off-on-start switch

b. A tachometer and combination hourmeter to continuously read engine speed

c. Engine lubricating oil pressure gage

d. Engine cooling liquid temperature gage

e. Automatic engine shutoff due to high coolant temperature with manual reset

f. Automatic engine shutoff due to low lubricating oil pressure with manual reset

g. Manual throttle control

h. Battery-charging ammeter

i. Engine priming system control

j. Pump discharge pressure gage (4-1/2-inch dial and 0 to 160 psig scale range).

3.7.8.1 All controls, gages, et cetera, shall be shock-mounted and provided with surge dampeners as necessary to provide stability of the indicating needle, provide proper and consistent functioning of the controls, and prevent damage. The control box shall provide for easy access for repair, checking, and replacement.

3.7.8.2 Control box light. The control panel of the control box shall be provided with a light for nighttime operation. The light shall be provided with an on-off switch and shall be wired such that it will only light when the engine on-off-start switch is in the on position.

3.7.9 Ground rod. One type II, style I, 5-foot-long ground rod in accordance with MIL-R-11461 shall be provided. The ground terminal shall be bolted to the base

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forming an electrical bond with a maximum of 0.5 ohm resistance. Provisions shall be incorporated for storing and retaining the ground rod with cable attached.

* 3.7.10 Gate valve assembly. The gate valve assembly shall consist of a 4-inch aluminum gate valve in accordance with MIL-V-58039 with one 4-inch male and one 4-inch female cam-locking, quick-disconnect coupling half with a dust plug and dust cover and gasket in accordance with MIL-C-27487, class I. Security chains shall be provided and shall be 1/16-inch-diameter cable in accordance with MIL-C-5424, except with a clear nylon jacket.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Electromagnetic interference. The pumping assembly shall conform to the radio interference requirements of MIL-STD-826.

3.10 Acoustic noise levels. The acoustic noise levels produced by the pumping assembly shall not exceed the values specified in table I when measured as specified in 4.6.8.

TABLE I

Maximum Noise Level Limits

Octave Band Center Frequencies (Hz)	Sound Pressure Level (Decibels Ref. 0.0002 Dynes/CM^2)	
	Near-Field Positions	Operator's Position
63	115	110
125	110	104
250	102	97
500	95	92
1,000	92	88
2,000	88	86
4,000	88	85
8,000	85	85

3.11 Dimensions. The pumping assembly, less trailer, shall not exceed a length of 92-1/4 inches, width of 34 inches, nor a height of 60 inches.

3.12 Weight. The pumping assembly dry weight (less trailer) shall not exceed 2,000 pounds.

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3.13 Finishes and protective treatment

3.13.1 Cleaning, painting, plating, anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808. A listing of the procedures used shall be furnished the procuring activity and shall be submitted with the test report specified in 4.4.2.

3.13.2 Painting. Exposed metal surfaces shall be color 34079 of FED-STD-595.

3.13.3 Anodizing. All aluminum moving parts and parts subject to wear shall be given a hard anodic coating.

3.14 Operational markings

3.14.1 Data plates. Instruction and identification data shall be permanently and legibly etched upon metal plates in accordance with MIL-P-514. The plates shall be aluminum alloy with a minimum thickness of 18 gage (0.04). Component identification plates shall be aluminum alloy with a minimum thickness of 0.032 inch. All plates shall be anodized or etched to produce raised natural metallic colored markings with a dark color background.

3.14.2 Diagrams. Schematic diagrams of the electrical and controls systems shall be provided. Each switch, et cetera, on the diagram shall be properly identified to correspond to the markings on like parts on the pumping assembly.

3.14.3 Operating instructions. Operating instructions for the pumping assembly shall be clearly and concisely stated with reference to the actual accessories giving the identification as indicated on the diagrams. Operations which affect safety of operation of the pumping assembly shall be listed at the top of the plate under the title CAUTION which shall appear in larger print or otherwise be designed to stand out.

3.14.4 Transportation data plate. A transportation data plate containing at least the following information shall be mounted on the tow bar:

a. Shipping weight (This shall be the weight of the assembly after being used and then drained as completely as possible)

b. Loading cubage

c. Overall height, width, and length

d. Horizontal and vertical location of the center-of-gravity.

3.14.5 Gages and controls. Gages and controls shall be marked indicating safe operating range and red level at unsafe conditions.

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3.15 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.16 Workmanship. The pumping assembly, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting: alignment of parts, and tightness of assembly screws and bolts; et cetera.

3.16.1 Dimensional tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the pumping assembly, they shall be held or limited accordingly.

3.16.2 Screw assemblies. Unless torque is specified, assembly screws and bolts shall be tight.

3.16.3 Riveting. Riveting operations shall be carefully performed to insure tight and satisfactory heading of rivets.

3.16.4 Gear and lever systems. Gear and lever systems shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, and other irregularities. When required for accurate adjustments, gear assemblies shall be free of backlash.

3.16.5 Cleaning. The pumping assembly shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slugs; stray bolts and nuts; pipe compound; rust; metal particles; and other foreign matter shall be removed during and after final assembly.

3.16.5.1 Weld beads. Weld beads shall be thoroughly cleaned, and spattering shall be removed from the pumping assembly.

3.16.6 Piping. The installation of piping, tubing, wiring, et cetera, shall be accomplished in such a manner as to prevent rubbing and wear due to vibration. The tubing and plumbing shall be assembled to permit adequate flexibility under torsional forces induced during operation and transportation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the pumping assembly shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Test fluid. Unless otherwise specified herein or in the detail component specifications, the test fluid shall be grade JP-4 fuel conforming to MIL-T-5624.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. Unless otherwise specified (see 6.2), one pumping assembly shall be subjected to the preproduction tests specified under 4.6. Additional assemblies could be provided for the environmental and mobility tests if a shorter preproduction test time is required.

4.4.2 Test report. Upon completion of the preproduction tests, a preproduction test report shall be prepared in accordance with MIL-STD-831. An authorized Government quality control representative shall countersign all data sheets and the report. The following shall be incorporated as part of an enclosure to the test report:

a. 35-millimeter colored slides - left front three-quarter view and right rear three-quarter view as well as other slides of the assembly with test apparatus

b. List of all components and items noting the manufacturer's name, part or model number, applicable specification, conformance or nonconformance to the specification (if nonconformance, a statement as to where and the reason for shall be included, metals used in the fabrication of all items that come into contact with the fuel)

c. Certification by component manufacturers that elastomers which come into contact with the fuel are compatible with fuels conforming to MIL-T-5624 and MIL-G-5572

d. QPL listing, certified test report, or Government agency approval letter (as applicable) for the components and items being used on the pumping assembly

e. A listing of all cleaning and finishing procedures in accordance with MIL-STD-808 to be used on the production pumping assemblies and all components.

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4.4.2.1 Reliability and maintainability information. The following information shall be included as an attachment accompanying the preproduction test report (same quantity and quality of copies) or shall be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made and behind the predicted effectiveness of these corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each pumping assembly shall be subjected to the following tests described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Mechanical operation . . . . . See 4.6.2

c. Static pressure tests . . . . . See 4.6.3

d. Flow tests . . . . . See 4.6.4.1 (at only 40 and 100 percent rpm).

4.6 Test methods

4.6.1 Examination of product. The pumping assembly shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to compliance with design requirements on preproduction assemblies. Particular attention shall be given to materials, workmanship, dimensions, protective finishes and their applications, welding, riveting, and markings on the pumping assembly. The pumping assembly shall be examined to assure protection is provided for devices, components, et cetera, that could otherwise easily be inadvertently damaged.

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4.6.2 Mechanical operation. A thorough check shall be made of the pumping assembly to insure proper functioning of all mechanical parts, accessibility of all controls and levers, interchangeability of removable parts, and other checks that will insure every degree of safety and performance.

4.6.3 Static pressure tests. The pumping assembly shall be subjected to a hydrostatic pressure of 200 psig. This pressure, once attained and the pressure producing pump is discon- ted or valved off, shall be maintained for a minimum period of 30 minutes. The initial and final pressures shall be recorded and included in the test report. Evidence of leakage or impending failure shall be cause for rejection. The pump housing shall be subjected to the hydrostatic pressure specified in 3.7.4.1, if greater than 200 psig.

4.6.4 Operation tests. The pumping assembly shall be subjected to the tests specified in 4.6.4.1. During these tests, the following data shall be recorded:

a. Date

b. Time started

c. Time finished

d. Ambient temperature

e. Fuel temperature (supply)

f. Pump suction pressure

g. Pump discharge pressure

h. Flow, gpm

i. Total flow, gallons

j. Engine rpm

k. Engine oil pressure

l. Engine coolant motor temperature

m. Engine operational time

n. Engine starts.

4.6.4.1 Flow tests. A minimum of five pump head capacity curves at constant rpm shall be obtained. Each curve shall have a minimum of four test points with one at a closed discharge, one against a low discharge pressure, and the other two equally

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spaced between the above two points. The rpm shall be at 32, 52, 68, 84, and 100 percent of the rpm to produce the 600 gpm at a total head of 130 psig. The pumping assembly shall have no instability or adverse vibration during these flow tests. The pumping assembly shall be operated at each point a minimum of 15 minutes. The tests shall be conducted with a static suction lift of 10 feet. The suction characteristics of the pump (net positive suction head) at 100 percent rpm shall be obtained at flow rates of 400, 500, and 600 gpm.

4.6.5 Dry operation test. The pumping assembly, pump, and piping shall be completely drained. Without introducing liquid into the piping or pump, the pump shall be operated at maximum rated rpm for 5 minutes. There shall be no appreciable wear, damage, overheating, or other failure as a result of this test.

4.6.6 Priming test. The pump housing shall be filled with sufficient fluid to effect a priming condition (not more than 5 gallons), and the pump shall demonstrate its ability to eliminate the air from a 6-inch-diameter by 10-foot-long hose and establish a positive pressure of at least 20 psi at the pump discharge.

4.6.7 Electromagnetic interference tests. The pumping assembly shall be tested for radiated interference in accordance with method 7001 of MIL-STD-826.

*4.6.8 Acoustic noise level test. The acoustic noise levels produced by the pumping assembly shall not exceed the values specified in 3.10 in any octave band at any of the measurement positions specified under 4.6.8.1. All acoustic measurements shall be made during rated capacity operation of the unit. Ambient background noise levels at all measurement positions should be at least 10 db below the octave band sound pressure levels produced by the pumping assembly. The acoustic measurement instrumentation shall be as specified in 4.6.8.2.

*4.6.8.1 Measurement positions

*4.6.8.1.1 Near-field positions. Measurements shall be made at eight positions equally spaced every 45 degrees on a 10-foot radius circle centered on the geometrical center of the pumping assembly. The microphone shall be located 5 feet above the pavement level and oriented toward the center of the unit.

*4.6.8.1.2 Operator's position. Measurements shall be made at the operator's position at the control panel, with the microphone 5 feet above the pavement level.

*4.6.8.2 Instrumentation. The basic instrumentation shall consist of a sound level meter in conjunction with an octave band analyzer. This instrumentation shall meet the requirements specified in USA Standards S1.4 and Z24.10, except the octave band analyzer shall employ the preferred frequencies specified in USA Standard S1.6. An up-to-date calibration and frequency response of the entire measurement system, including microphone, shall be available to permit correction of the measured data. The range of frequency response of the system shall be from 45 to 11,200 Hz and the response stated within ±2 db. An acoustical calibrator shall be used for an

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end-to-end system calibration from the microphone to the meter. The sound pressure level (in db ref. 0.0002 dynes/cm^2) produced by the calibrator, when used in conjunction with the particular microphone in the system, shall be known within ±0.5 db.

4.6.9 Environmental tests. The following environmental tests shall be conducted on the pumping assembly in accordance with the specified methods of MIL-STD-810 without the cover required in 3.7.10. Unless otherwise specified (see 6.2), an engineering evaluation may be performed in lieu of the tests specified in 4.6.9.3 through 4.6.9.6. The engineering evaluation shall consist of a written detailed analysis of the pumping assembly and all its parts and components to satisfactorily accomplish the required environmental conditions.

4.6.9.1 High temperature test. The pumping assembly shall be subjected to high temperature in accordance with method 501, procedure I. No adverse effects that would affect operation at normal temperature shall result from this exposure. The chamber temperature shall be reduced to 125°F, and the test specified in 4.6.4.1 shall be conducted at 100 percent rpm The engine shall then be stopped for 5 minutes and restarted. The pumping assembly shall be operated at 600 gpm with a total head of 130 psig for 1/2 hour. The engine shall then be stopped for 5 minutes and restarted. The engine shall be operated at idle speed for 1/2 hour. All engine starts shall be accomplished within 5 minutes of the original attempt. The pumping assembly shall operate satisfactorily and perform its intended functions throughout this test. There shall be no leakage, damage, evidence of thermal degradation, or impending failure of any portion of the assembly. The engine coolant temperature shall not exceed that recommended by the manufacturer or 10 degrees below the boiling point rating of the system, whichever is lower and there shall be no loss of coolant. The engine shall be restarted without auxiliary cooling or electrical power.

4.6.9.2 Low temperature test. The pumping assembly shall be subjected to low temperature in accordance with method 502, procedure I, except after the -80°F soak the chamber temperature shall be raised to -20°F and stabilized for 24 hours. The batteries shall be maintained at a temperature of 0° -0° +10°F. The engine shall then be started within 10 minutes of the initial attempt and shall demonstrate the ability to pump fuel at 300 gpm for a minimum period of 10 minutes. The pumping assembly shall operate satisfactorily and perform its intended function. The pumping assembly shall be inspected during the test and there shall be no leakage, damage, evidence of degradation, failure, or impending failure of any portion of the assembly.

4.6.9.3 Humidity test. The pumping assembly shall be subjected to humidity in accordance with method 507, procedure I, and tested as follows:

*4.6.9.3.1 Operational test. After the performance check and inspection, the pumping assembly shall be operated for 48 hours producing a 600-gpm flow with a total head of 130 psi. The pumping assembly shall be started and stopped eight times, at equally spaced intervals, during the 48-hour period.

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*4.6.9.4 Fungus test. The pumping assembly shall be subjected to fungus in accordance with method 508, procedure I, and tested as spec fied in 4.6.9.3.1.

*4.6.9.5 Salt-fog test. The pumping assembly shall be subjected to salt-fog in accordance with method 509, procedure I, and tested as specified in 4.6.9.3.1.

*4.6.9.6 Dust test. The pumping assembly shall be subjected to sand and dust in accordance with method 510, procedure I, and tested as specified in 4.6.9.3.1.

*4.6.9.7 Rain. The pumping assembly shall be subjected to rain in accordance with method 506, procedure I, and tested as specified in 4.6.9.3.1.

4.6.10 Air transportability evaluation. A detailed engineering evaluation shall be conducted on the pumping assembly to determine compliance with the tests specified in MIL-A-8421 for air transport in Types C-123, C-124, C-130, C-133, and C-141 aircraft. The evaluation shall include all major components and their capability to withstand the 8-g forward load as well as the g loadings specified in MIL-A-8421. The evaluation shall include the tiedown fittings.

4.6.11 Weight. The complete pumping assembly shall be weighed both in the dry and drained conditions. In addition, the weight of each component shall be determined, recorded, and included in the test report.

*4.6.12 Mobility tests. While in a drained condition, the pumping assembly shall be towed by a truck tractor or a vehicle having a pintle height of 10 to 30 inches over paved highway at an average speed of 20 mph for a distance of 35 miles, over gravel roads at an average speed of 10 mph for a distance of 15 miles, and over Belgian block at an average speed of 8 mph for a distance of 35 miles. At least 10-short-radius right turns, 10 short-radius left turns, and 5 sudden stops shall be conducted. Throughout the test, the pumping assembly shall be observed as to its trackability, ease of handling, tendency to yaw, tendency to sway, et cetera. Upon completion of the tests, the pumping assembly shall be inspected for misalignment, distortion, or other damage sustained as a result of the testing. Undesirable roadability features and any damage resulting from the test shall be cause for rejection. A complete description of any adverse handling characteristics or damage sustained shall be recorded on the data sheets. At the conclusion of the mobility tests, the test specified in 4.6.4.1 at 100 percent rated rpm shall be successfully conducted.

4.6.13 Endurance test. The pumping assembly shall be subjected to the following 345-hour endurance test. The pumping assembly shall show no indication of impending failure or degradation of components as a result of this test.

600 gpm flow with a total head of 130 psig for 48 hours

450 gpm flow with a total head of at least 100 psig for 96 hours

300 gpm flow with a total head of at least 80 to 90 psig for 96 hours

100 gpm flow with a total head of at least 60 to 70 psig for 57 hours

600 gpm flow with a total head of 130 psig for 48 hours

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*4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

*5.1 Preparation for delivery shall be in accordance with MIL-E-10062.

6. NOTES

6.1 Intended use. The A/M32R-22 trailer-mounted 600-gpm pumping assembly is intended for use as an alternate to those provided with the air transportable hydrant refueling system.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When components are to be retained by other than removable-type fasteners such as nuts and bolts (see 3.7.2)

c. Location and conditions for preproduction testing (see 4.4)

d. When more than one pumping assembly is to be subjected to the preproduction tests (see 4.4.1)

e. When an engineering evaluation will not be accepted in lieu of the testing specified in 4.6.9.3 through 4.6.9.6.

6.3 Definitions. For the purpose of this specification, the following definitions shall apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-X} \text{ where } X = \frac{\text{Mission time}}{\text{MTBF}}$$

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The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees of freedom) distribution even though the failure rate for the test time is zero.

6.3.1.1 Mean-time-between-failures. The mean life or mean-time-between-failures is defined as the arithmetical mean (average) of the operating time between failures.

6.3.1.2 Mission time. Mission time is defined as the expected or average operating time for each use of the equipment.

6.3.2 Highway. A highway is defined as a road of level concrete or macadam.

6.3.3 Belgian-block pavement. Belgian-block pavement is defined as pavement consisting of a very rough wavy, hard surface unaffected by changes in climatic conditions.

6.3.4 Self-priming centrifugal pump. A self-priming centrifugal pump is defined as one designed to have the ability to prime itself automatically, after being initially filled, when operating under a suction lift, and to free itself of entrained gas without losing its prime and to continue normal pumping of without attention. Use of an ejector to effect priming will not be acceptable.

6.4 Test courses. The test courses in existence at Aberdeen Proving Ground, Aberdeen, Maryland, on the date of invitation for bids will be considered the standard test courses. Should the test be conducted at an alternate test course, the contractor is still responsible for designing and constructing the vehicle to successfully perform the tests on the standard test courses at Aberdeen Proving Ground.

*6.5 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

amp . . . . . ampere

db . . . . . decibels

g . . . . . gravitational units

gpm . . . . . gallons per minute

Hz . . . . . Hertz, cycles per second

mph . . . . . miles per hour

MTBF . . . . . mean-time-between-failure

psi . . . . . pounds per square inch

psig . . . . . pounds per square inch gage

rpm . . . . . revolutions per minute.

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6.6 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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overhaul The steam cleaner is used in fixed shops as well as in the field

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